Ruchir Jadhav

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WORK EXPERIENCE

Easley Dunn Productions (Mobile Software Studio) Remote
Software Development Engineer *Feb 2026-Present*

- Authored 67 of 119 merged pull requests (422 commits) on a ~15-contributor team, assigning tasks and making implementation decisions as de facto backend lead for SafetyNet, a React Native app on Node.js/Express and PostgreSQL.
- Hardened the PostgreSQL layer (13 tables, 24 migrations) by adding session revocation, transactional multi-step writes, atomic ON CONFLICT upserts, and a fix for full-table scans on phone lookups, verified with before/after EXPLAIN.
- Raised release confidence with 453 backend integration tests on a real Postgres database and 7 required CI checks per PR, including a migration-safety guard that blocks destructive migrations.
- Built the deploy pipeline: auto-deploy to an internal environment with health checks and automatic rollback, one-click promotion to production, and a production move to a Google Cloud VM with twice-daily backups and a passed restore drill

Amazon Web Services (AWS) Seattle, WA
Software Development Engineer Intern *May 2025-Aug 2025*

- Built a four-component backfill system in Go for the Amazon EKS Metrics Service, resolving CloudWatch API outages that had caused up to 6-hour metric data gaps; implemented failure detection, hierarchical data preservation, and chronological replay for zero data loss.
- Designed a hot-reloadable failure simulation framework with per-log-group injection rates (0–100%) and S3-synced JSON config for safe outage testing in beta; validated recovery end-to-end via before/after CloudWatch dashboard comparisons.
- Coordinated recovery by implementing offset-timed reprocess (30 s) and republish (45 s) services with AWS-specific retry/backoff, CloudWatch Logs batching (100 events/10 KB), and S3 lifecycle tiering (Standard→IA→Glacier), which shortened outage recovery time and ensured continuous data availability during extended incidents.

BusinessNext (Financial Services CRM for Banks and Credit Unions) Mumbai, India
Data Engineer *Sep 2023-May 2024*

- Built and refined SQL queries on Linux using Django ORM to power client-portfolio dashboards for Relationship Advisors in BusinessNext's banking CRM, enabling the frontend React Native team to deliver responsive visual reporting interfaces.
- Developed data logic to surface product recommendation opportunities for banking clients, translating Relationship Advisor requirements into structured queries that identified better-fit financial offers from existing account data.

ACADEMIC PROJECTS

FACTUAL — Real-Time Multimodal Fact-Checking Jan 2025-May 2025

- Co-designed an end-to-end fact-checking pipeline combining OpenAI Whisper ASR, transformer-based claim extraction, and a Retrieval-Augmented Generation (RAG) module with Google Search/News API retrieval and a self-critique loop where a judge LLM iteratively refines queries (up to 10 iterations) to reduce hallucination.
- Built a custom 400-claim benchmark spanning pre- and post-model-cutoff claims, augmented with speech-like disfluencies and converted to audio; evaluated GPT-4o and Gemini across zero-shot, RAG, and agentic configs, with the best (GPT-4o Agentic) reaching 84% accuracy and 0.83 F1 on post-cutoff claims.

Multimodal Emotion Detection Jan 2025-May 2025

- Balanced the MELD dataset across 7 emotion classes using audio/video augmentation; fine-tuned a LoRA-augmented transformer with OpenFace facial Action Unit features to improve classification on underrepresented classes.
- Built an end-to-end multimodal inference pipeline combining BERT (text), HuBERT (audio), and video modules with automated feature extraction and late fusion, achieving ~70% weighted accuracy across cross-validation folds.

EDUCATION

University of Southern California Los Angeles
Master of Science in Computer Science Jun 2024-Dec 2025

University of Mumbai Mumbai
Bachelor of Technology in Computer Engineering Jul 2019-Apr 2023

LANGUAGES & TECHNOLOGIES

Languages: Python, Go, JavaScript, TypeScript, SQL
Backend: Node.js, Express, REST API design, Socket.IO (WebSockets), JWT authentication, Firebase Auth, bcrypt, rate limiting, AES-256-GCM encryption
Cloud and infrastructure: AWS (CloudWatch, S3, EKS), Google Cloud (Compute Engine, Cloud Storage, Error Reporting, IAP), Firebase Analytics, Cloudflare Tunnel, Tailscale, Linux
Mobile and frontend: React Native, React, TypeScript, Leaflet
Databases: PostgreSQL, parameterized SQL (pg), recursive CTEs, schema migrations (node-pg-migrate), indexing and EXPLAIN query tuning, transactions
Analytics: Firebase Analytics, GA4, BigQuery, Looker Studio
Testing and CI/CD: Jest, Supertest, integration testing, contract testing, React Native Testing Library, GitHub Actions, automated rollback, migration-safety checks